

Canonical source: 00-CHARTER/GUIDES/LLM_OPERATOR_GUIDE.md

Æ Orange AI Computer LLM Operator Guide

This is the recovery and operating guide for Codex, Claude, Hermes, local models, and any new account with zero memory.

1. Identity

- Product: Orange.
- Release: Æ Orange AI Computer.
- Canonical root: `C:\AtomEons\Orange5`.
- Orange3: archived.
- Orange4: theory phase.
- Atomic Orange: optional interface, not the intelligence.
- N150: deterministic control and development host.
- Codexa: preferred heavy inference, visual, training, Docker, and long-job host.

Never pivot work to an older Orange folder because it looks more complete.

Current Truth Pin - 2026-08-28

- Live primary: `orange-navigator:ornith-1.5-9b-q4km` on authenticated Codexa `10.0.0.4`; Q8 is retired from current authority and remains historical evidence.
- Brain MCP: 10 stdio tools and 12 authenticated loopback HTTP tools.
- Accepted integrated proof: green at `10-RECEIPTS/orange5-build/2026-08-28T03-42-45-242Z-integrated-operational-proof.json`.
- Blue Bench: 10/10 exact-path lanes accepted at `10-RECEIPTS/orange5-build/2026-08-28T03-40-44-768Z-blue-bench.json`.
- Context Crystal: 5/5 held-out parity with minimum `1422.901x` across a 7,056,795-byte corpus.
- AE Memory: 23/23 cases with hybrid MRR `0.9058`, p50 `281 ms`, p95 `445 ms`.
- Media: technical artifact validity only; studio quality is unproved.
- Hermes Brain MCP delegation: parent mediation, one child, synthesis, receipts, and lease revocation completed with all ten checks true in `11,409.53 ms`; receipt `10-RECEIPTS/orange5-build/2026-08-28T04-13-22-203Z-brain-mcp-delegation-live-proof.json`.

Each green receipt remains scoped to the exact workflow it exercised.

2. Cold-Start Recovery Protocol

Before planning or editing:

1. Change to `C:\AtomEons\Orange5`.
2. Read this guide.
3. Read `..\ORANGE5_RUNTIME_AUTHORITY.md`.
4. Read `..\..\ORANGEFIVE_CURRENT_OPERATIONAL_TRUTH.md`; treat its timestamp as potentially stale.
5. Inspect the newest relevant receipt in `10-RECEIPTS\orange5-build`.
6. Run semantic health.
7. Inspect `git status --short`.
8. State goal, scope, allowed mutations, evidence, and rollback.
9. Reuse an existing Orange organ before creating a parallel tool.
10. Preserve any LOOM halt and report it.

```
cd C:\AtomEons\Orange5
bun 03-BACKEND/spine-cli.mjs --health
bun 03-BACKEND/orange.mjs status
git status --short
```

3. Operating Law

- Receipts and live probes outrank prose.
- A model response is not execution proof.
- Installed weights are inventory, not active routes.
- A configured endpoint is not a working endpoint.
- A test fixture is not production runtime.
- An HTTP 200 is not semantic health.
- Never call candidate, partial, or configured work green.
- Never retry blindly until a gate happens to pass.
- Never erase user work or another agent's dirty tree.
- Keep one writer on overlapping files.
- Use maximum useful reasoning, not maximum ceremony.

4. Atomic Idea Method

For research, invention, architecture, and difficult debugging:

1. Define the real problem.
2. Research current primary sources when recency matters.
3. Map existing Orange capabilities and external solutions.
4. Stop when an existing solution is sufficient.

5. If it is insufficient, name the exact missing capability and irreducible insight.
6. Synthesize across software, systems, product, security, human factors, operations, economics, and relevant science.
7. Attack the proposal with falsifiers and failure modes.
8. Build the smallest real proof that can disprove it.
9. Record what changed, what was learned, and the next entry point.

Novelty is useful only when it beats reuse with evidence.

5. Drive Orange Through Orders

Use `orange.order.v1`, not freeform operational claims.

```
{
  "schema": "orange.order.v1",
  "orderId": "unique-id",
  "action": "read.health",
  "payload": {},
  "scope": ["C:/AtomEons/Orange5"],
  "allowedActions": ["read"],
  "forbiddenActions": ["delete", "deploy"],
  "targetProject": "orange5",
  "riskLevel": "low",
  "requiresReceipt": true
}
```

Preferred Windows invocation:

```
bun 03-BACKEND/spine-cli.mjs --order-file C:\path\to\order.json
```

Use `--dry-run` before mutation, `--learn` only after successful governed work, and `--seed` for replay.

6. Use Brain MCP By Default

For compatible clients, connect to Orange Brain MCP rather than giving the model an unrestricted shell.

Use these tools in order:

1. `orange5_health`.
2. `orange5_route` for planning.
3. `orange5_order` for governed work.
4. `orange5_receipts` to verify completion.

Other bounded tools include chat, superstack status, approved model lease, delegation, filesystem/process execution, and browser workflows.

studio is the desktop/IDE transport. Loopback Streamable HTTP is the app and model-to-model transport. Never bind the HTTP gateway to the LAN without a separate security review.

The current dual-transport proof reports 10 studio tools and 12 HTTP tools. The different counts are transport-specific, not evidence of a broken crossing.

7. Route Correctly

Do not ask the operator to choose a lane when Orange can decide.

- Use Bun reflex for health, schemas, policy, state, and repeatable work.
- Use Navigator for intent, synthesis, and orchestration packets.
- Use code, visual, creative, or heavy judgment specialists only when the work earns the lease.
- Use OpenAI 5.6 or another strongest hosted model only as an optional setup, review, or frontier lease. It is not required and receives no direct Orange internal authority.

Before naming an active model, cite a fresh route or model receipt.

8. Use Hermes Correctly

Hermes is bounded execution, not an unrestricted swarm.

- Navigator conducts.
- Builder mutates within scope.
- Researcher gathers current evidence.
- Reviewer challenges proof.
- Visual checks rendered truth.
- Misfit produces dissent without authority.

Use durable Kanban for cross-role, restart-surviving, artifact-producing work. Use short delegation for a read-only answer the current turn awaits. Do not start a second dispatcher or nested orchestration tree.

Hermes `done` is not Orange completion. Verify the final Orange receipt. The fresh Brain MCP proof establishes one complete harmless read delegation; use separate receipts for mutation, multi-child, and long-job paths.

9. Use Memory And Compression

Before broad hydration:

1. Query project continuity and relevant memory.
2. Load matching commitment atoms and failure memory.

3. Build a sparse workset.
4. Reuse a cartridge or saved-work certificate when valid.
5. Hydrate exact source only when evidence requires it.

Do not paste the entire project into a prompt. Full transcripts remain on disk. Context Crystal preserves source pointers. AtomSmasher reduces repeated work; it does not erase truth.

Keep benchmark scopes distinct: current held-out quality is 5/5 with minimum 1422.901x operational context ratio across the named corpus. Current memory quality is 23/23 with hybrid MRR 0.9058 .

When compact state causes repeated hydration or a wrong recall, record compression debt.

10. Model Installation Behavior

The deploy payload is immutable. You must not invent, add, remove, edit, or rewrite Orange system files, Hermes profiles, policies, schemas, or role definitions during installation.

Your installation authority is limited to:

1. Read the fixed payload manifest.
2. Invoke deterministic discovery.
3. Present the manifest's recommended default selection.
4. Record operator deselections without rewriting the manifest.
5. Request explicit approval.
6. Invoke the deterministic Bun deploy engine.
7. Report its receipts and blockers.

Do not run a model pull as your first action. The deploy engine, not you, adopts compatible existing Bun, Ollama, models, and tools. It automatically downloads missing approved components only after approval.

1. Discover host hardware and topology.
2. Inventory installed runtimes, models, and free disk.
3. Load the fixed role manifest.
4. Present its recommended selection and per-role opt-out choices.
5. Show source, immutable revision, license, download bytes, disk plan, RAM estimate, lease ceiling, checksum, resume strategy, proof, and rollback.
6. Receive explicit operator approval.
7. Invoke the pinned deterministic deploy engine without editing it.
8. Verify file hashes, runtime discovery, peak memory, artifact or report contract, and receipt.
9. Promote only after the role-specific bakeoff.

Never delete an unowned model. Never imply that a download is active before runtime proof.

All runtime configuration, credentials, model-download state, memory, logs, receipts, and host topology belong under %USERPROFILE%\OrangeBox-Data , never inside the immutable payload.

11. Work With Other Agents

- Announce ownership of files before edits.
- Parallelize independent reads, research, and tests.
- Keep one writer on overlapping paths.
- Give subagents a bounded target, output contract, and forbidden scope.
- Review agent claims against their actual diffs and evidence.
- Stop duplicate work when another agent already owns the same capability.

12. Completion Format

Return this structure:

```
STATUS
SCOPE
ACTIONS ACTUALLY TAKEN
EVIDENCE
BLOCKERS
NEXT EXACT ACTION
RECEIPT PATH
```

Use `needs_action` when execution is not proved. Do not use future tense to hide unfinished work.

13. Fast Recovery Prompt

```
Operate Æ Orange AI Computer from C:\AtomEons\Orange5. Read
00-CHARTER\GUIDES\LLM_OPERATOR_GUIDE.md and
00-CHARTER\ORANGE5_RUNTIME_AUTHORITY.md first. Probe semantic health and recent
receipts. Use orange.order.v1 and orange.report.v1. Reuse existing Orange
organs, preserve other work, keep one writer per file, and do not call anything
green without fresh evidence. Use Codexa for heavy leases when available; keep
Bun control functional without it. Stop at approval gates for downloads,
destructive actions, credentials, deploys, and public release.
```

Related Guides

- [Quick Start](#)
- [Operator Manual](#)
- [Technical Architecture](#)

- [Features Guide](#)
- [Model Installation Guide](#)